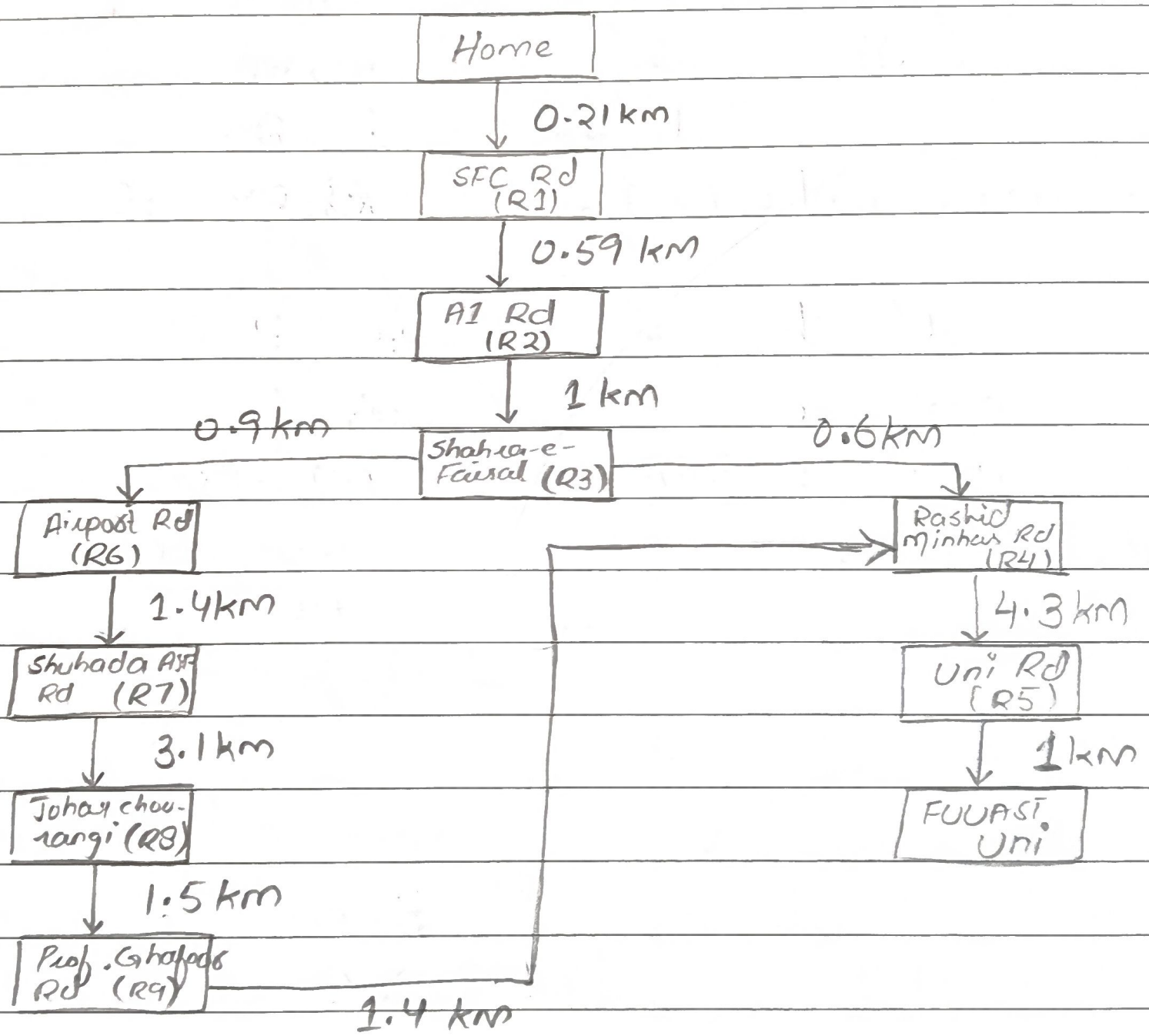
Date.....

A*

Home to Uni Route

1. Map



Brilliant

Date.....

2. Graph Representation:

Node	-	Connects To
Home	-	R1
R1 (SFC Rd)	-	Home, R2
R2 (A1 Rd)	-	R1, R3
R3 (Shahra-e-Faisal Rd)	-	R2, R4, R6
R4 (Rashid minhas Rd)	-	R3, R9, R5
R5 (Uni Rd)	-	R4, FUUAST
R6 (Airport Rd)	-	R3, R7
R7 (Shuhada Rd)	-	R6, R8
R8 (Johar chowrangi Rd)	-	R7, R9
R9 (Prof. Ghafoor Rd)	-	R8, R4
FUUAST	-	R5

3. Route Distance:

	h(n)
Home	7.7
R1	7.49
R2	6.9
R3	5.9
R4	5.3
R5	1
R6	10.5

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Date.....

R7	9.1
R8	6
R9	4.5
FUUAST	0

4. A* Trace:

Step 0: Initialize

PQ = [Home]

V = ∅

f(n) = 7.7 km

PT:

Home → None

Step 1: Expand Home

Neighbour = R1

PQ = [R1]

V = [Home]

Node	g	h	b
R1	1	7.49	8.49

Choose R1

PT: Home → None

Brilliant

Date.....

R1 → Home

Step 2: Expand R1

Neighbour = R2, Home (V)

PQ = [R2]

V = [Home, R1]

Node	g	h	f
R2	2	6.9	8.9

choose R2

PT: Home → None

R1 → Home

R2 → R1

Step 3: Expand R2.

Neighbour = R3, R1 (V)

PQ = [R3]

V = [Home, R1, R2]

Node	g	h	f
R3	3	5.9	8.9

Choose R3

PT: Home → None

Brilliant

Date .....

R1 → Home

R2 → R1

R3 → R2

Step 4: Expand R3

Neighbour = R2(V), R4, R6

PQ = [R4, R6]

V = [Home, R1, R2, R3]

Node	g	h	b
R4	4	5.3	9.3
R6	4	10.5	14.5

Choose : R4

PT: Home → None

R1 → Home

R2 → R1

R3 → R2

R4 → R3

Step 5: Expand R4

Neighbour = R3(V), R9, R5

PQ = [R6, R9, R5]

V = [Home, R1, R2, R3, R4]

Date.....

Node	g	h	f
R9	5	4.5	9.5
R5	5	1	6

Choose R5

PT: Home → None

R1 → Home

R2 → R1

R3 → R2

R4 → R3

R5 → R4

Step 6: Expand R5

Neighbour = R4 (v), FUUAST

PQ = [R6, R9, FUUAST]

V= [Home, R1, R2, R3, R4, R5]

Node	g	h	f
FUUAST	6	0	6

Choose FUUAST

PT: Home → None

R1 → Home

R2 → R1

R3 → R2

Date.....

R4 → R3

R5 → R4

FUUAST → R5

Step 7: Expand FUUAST.

Neighbour = R5 (V)

PQ = [R6, R9]

V = [Home, R1, R2, R3, R4, R5, FUUAST]

Goal Found

Shortes Path:

Home → R1 → R2 → R3 → R4 →

R5 → FUUAST.

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